

NORDSTROM RACK – TURNKEY PROTOTYPE

SECTION 033536 – POLISHED CONCRETE FINISHING

1. Basis of Design: CTS Cement Manufacturing Corp. (Garden Grove, CA; 800-929-3030) "Rapid Set Cement All" fast-setting, high-strength, multi-purpose, non-shrink grout.
2. Acceptable options from the polishing system manufacturers (subject to compliance with Contract Document requirements and Architect's approval of color, technical parameters, and conformance to design intent) will be considered.

2.3 EQUIPMENT

- A. Field Grinding and Polishing Equipment:
 1. A multiple-head, variable-speed machine equipped with planetary, counter-rotating concrete grinding heads; ride on or walk behind machines, of various size and weights, with bonded abrasive pads affixed to heads for grinding concrete.
 2. Equipped with heavy-duty industrial HEPA filtration vacuum to capture and prevent concrete dust from escaping into interior spaces. Provide mandatory pre-separator for all metal bond cuts when grinding dry.
 3. Grinding and polishing equipment shall generate minimum 600 lbs head pressure.
- B. Edge Grinding and Polishing Equipment: Hand-held or walk-behind machines which produces same results, without noticeable differences as field grinding and polishing equipment, designed to access areas adjacent to vertical surfaces and inside corners.
- C. Burnishing Equipment: High speed walk-behind machines capable of generating 1000 to 2000 revolutions per minute and with sufficient head pressures of not less than 20 pounds (9 kg) to raise floor temperature by 20 deg F (-7 deg C).
- D. Diamond Tooling: Abrasive tools that contain industrial grade diamonds within a bonded matrix (such as metallic, resinous, and ceramic) attached to rotating heads to refine the concrete substrate.
 1. Allowable Grinding/Polishing Diamonds (or approved equivalents):
 - a. Metal Bonded Diamond Grit Sizes: 40, 80 and 150.
 - b. Hybrid Style Diamond: 30, 50, 100 or approved equivalent.
 - c. Resin Bonded, Phenolic Diamond Grit Sizes: 100, 200, 400, and 800, (or manufacturer's equivalent designations) as needed to achieve specified gloss.
 2. Progression of all polishing steps shall include sequential grit sizes that are not greater than twice the previous grit used.

2.4 ACCESSORIES

- A. Neutralizing Agent: Tri-sodium phosphate or baking soda.
- B. Water: Clear and potable.
- C. Kraft Curing Paper: Conforming to ASTM C171, Type 1.1.1, non-staining, moisture retentive, as specified by the system manufacturer.
- D. Semi-Rigid Joint Sealant: Rapid setting, two-component polyurea polymer liquid of 100 percent solids content with a minimum hardness of Shore A 75 (ASTM D2240) as recommended by the polishing system manufacturer; colored to match the natural gray concrete where no dye is applied.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Prior to starting work, carefully inspect installed work of other trades and verify that such work is complete to the point where work of this Section may properly commence. Notify the Architect in writing of conditions detrimental to the proper and timely completion of the work.